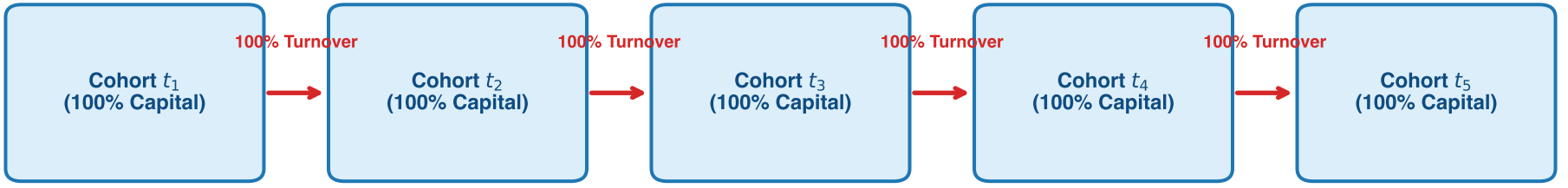
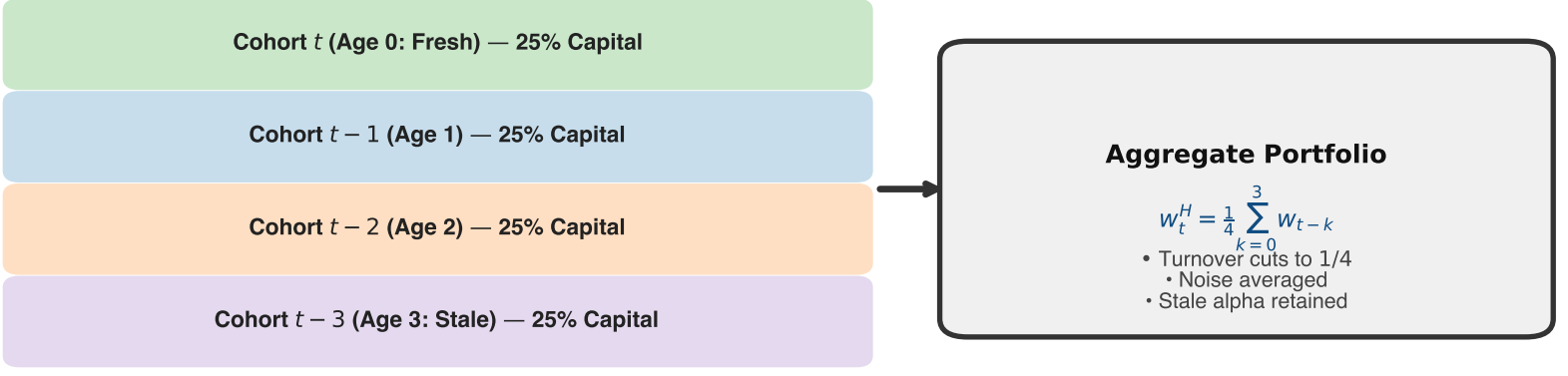
(a) Single – Period Execution ($H = 1$): High Responsiveness, Severe Rebalancing Churn



- 100% capital rebalanced every period on fresh forecast
- Captures peak initial signal, but incurs maximal turnover drag and estimation noise

(b) Overlapping Cohort Stacking ($H = 4$): Staggered Rolling, Mechanical Regularization



(c) The Triad of Competing Economic Forces Governing Optimal Horizon H^*

1. Alpha Dilution

$\partial \text{Alpha}/\partial H \leq 0$

Decaying signals in older cohorts drag down gross returns

Favors Shorter H ($H = 1$)

2. Cost Amortization

$\partial \text{Cost}/\partial H < 0$

Rolling $1/H$ capital each period amortizes rebalancing friction

Favors Longer H ($H \geq 4$)

3. Temporal Diversification

$\partial \text{Risk}/\partial H \leq 0$

Overlapping entry schedules cancel transient estimation noise

Favors Longer H ($H \geq 4$)